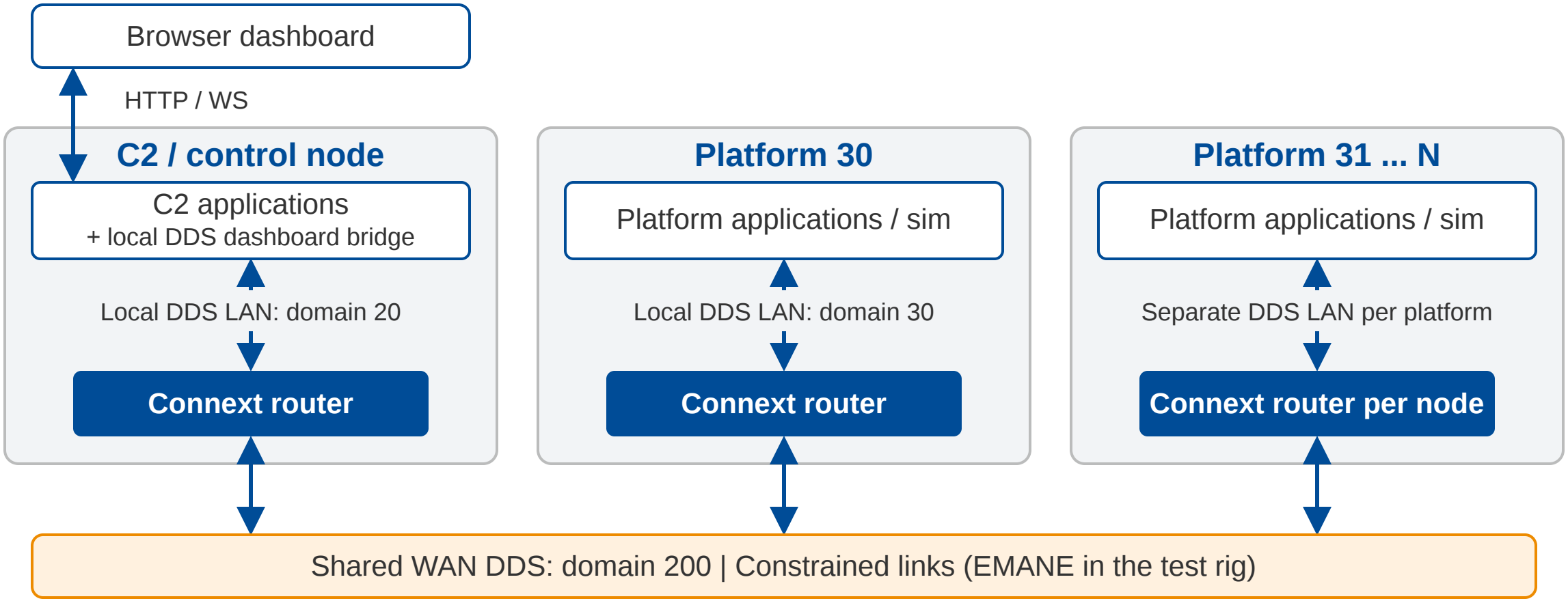
Current system



One router per node: selected commands, status, health, and team data cross the WAN.

Isolation mechanisms

Mechanism	Network	Discovery	Logical	Runtime changeable	Network bandwidth impact
VLANs / firewall rules	Network paths	—	—	Network admin	Limits discovery + data reach
Domain ID	Port ranges	Participant + endpoint	—	—	Scopes discovery + data
Domain tag	—	Participant + endpoint	—	—	Less endpoint discovery + data
DomainParticipant partitions	—	Participant + endpoint	Subsystem / team	Application	Less endpoint discovery + data
Publisher/Subscriber partitions	—	—	Groups of topics	Application	Less data; discovery remains
Topics	—	—	Topic	—	Discovery per endpoint; data per match
Content filters	—	—	Sample	Application	Less data if sender-filtered
Union discriminator	—	—	Sample (variant)	Application	Fewer endpoints: less discovery

Runtime changeable: no DDS entity recreation.
Bandwidth: endpoint count drives discovery; sender-side selection reduces user data.

Two strategies for data filtering

	Control exposure	Control interest
Driver	Source controls data availability	User controls scope / filter of available data
Control	Enable additional routes or reporting topics	Expand subscriptions or content-filter criteria
Example	Enable routes based on mission state	Request data based on teams or location
Benefit	Explicit control over what may leave a node	Each reader selects what it needs
Trade-off	Broader exposure can increase WAN traffic	WAN savings require interest to be applied upstream, not just after receipt

Use both: routes define what is available; reader interest selects within that boundary.

A reader request does not, by itself, enable a disabled route.

Discovery options

Option	How peers are found	Benefit	Trade-off
Multicast	Announce to a shared multicast group	No peer address roster	Requires multicast-capable network
Unicast: defined peers	Contact configured discovery addresses	No multicast or central service required	Maintain peer addresses as deployment changes
CDS	Cloud Discovery Service forwards participant announcements	Dynamic rendezvous without a custom directory	Operate a reachable discovery service
Unicast: bootstrap	A directory (e.g. C2) supplies peers; application adds them at runtime	Start with a bootstrap address, not the full roster	Application owns directory updates and recovery

Bootstrap: knowing one ordinary participant does not automatically discover its peers.

DDIL link-health measurement options

Measurement	Simple approach	What it tells us
Peer presence	Track existing health heartbeats	Is the peer reachable?
Round-trip time (RTT)	Acknowledge a heartbeat in the next heartbeat; subtract the reporting wait	How long is the round trip?
DDS protocol counters	Track per-peer repairs and unacknowledged data	Is reliable delivery under pressure?

Reuse existing traffic. Show how fresh each measurement is.

Presence, delay, and delivery pressure are separate facts, not a single health score.